[Letterhead]

[Date]

Dr. [Editor Name]
Physical Review X
American Physical Society
1 Physics Ellipse
College Park, MD 20740

RE: Manuscript Submission — "Fractal-Enhanced Topological Quantum Computing: Hilbert Space Scaling and Decoherence Suppression via Non-Semisimple TQFTs and Photonic Band Engineering"

Dear Editor,

I am pleased to submit the enclosed manuscript for consideration in Physical Review X. This work addresses a fundamental challenge in quantum computing: the resource overhead required for fault-tolerant operation. We present Aurphyx, an architecture that achieves a projected 16× reduction in error correction overhead through the synergistic combination of fractal geometry, non-semisimple topological quantum field theories, and photonic band engineering.

Summary of Key Results

1. Fractal Hilbert Space Scaling (Theorem 1) We prove that qubits arranged on Sierpiński gaskets (Hausdorff dimension $D_f = 1.585$) access exponentially larger Hilbert spaces than Euclidean lattices:

$$\dim(H_{acc}) = 2^{n \cdot D_f^{\alpha(k)}}$$

At $n = 12$ qubits and recursion depth $k = 3$, this yields a $10^4\times$ advantage. The result is validated by Qiskit simulations showing 7.1× advantage at $n = 5$.

2. Non-Semisimple Topological Universality (Theorem 2) We establish that non-semisimple modular tensor categories, via neglecton braiding (anyons with quantum dimension $d_\omega = 0$ but non-trivial braiding statistics), achieve universal quantum computation without magic-state distillation. This reduces gate overhead by approximately 100× compared to surface-code approaches.

3. Photonic Decoherence Suppression Hexagonal C_{6v}-symmetric photonic lattices provide:

- 21% band gaps (largest among topological photonic platforms)
- Anderson localization in flatband regions
- Decoherence suppression: $\gamma_{fractal}/\gamma_{Euclidean} = 0.063$ (16× improvement)

4. Experimental Validation Framework We propose four concrete protocols targeting NV-diamond coherence, trapped-ion Hilbert scaling, photonic band characterization, and Majorana T-junction fidelity, with combined budget $1.25M and 18-month timeline to Technology Readiness Level 4.

Significance and Fit for PRX

This manuscript is appropriate for Physical Review X because:

1. **Broad Impact**: The 16× overhead reduction addresses a central bottleneck in quantum computing, potentially advancing practical quantum advantage by 5–10 years.
2. **Interdisciplinary Synthesis**: The work bridges fractal mathematics, topological quantum field theory, photonic crystals, and experimental quantum physics—exemplifying the cross-disciplinary research PRX values.
3. **Rigorous Theory + Experimental Pathway**: We provide both formal theorems (Hilbert scaling, neglecton universality) and detailed experimental protocols with hardware budgets, success criteria, and risk mitigation.
4. **Timeliness**: Following Microsoft's Majorana-1 demonstration (99% Z-parity fidelity, 2025), topological quantum computing has entered a new era. Our work extends this foundation with fractal enhancement and non-semisimple TQFT integration.

Suggested Reviewers

We suggest the following experts as potential reviewers:

1. **Prof. Charles Marcus** (Niels Bohr Institute) — Topological quantum computing, Majorana nanowires
2. **Prof. Mikhail Lukin** (Harvard University) — NV centers, quantum networks
3. **Prof. Marin Soljačić** (MIT) — Photonic crystals, topological photonics
4. **Prof. Eric Rowell** (Texas A&M) — Non-semisimple TQFTs, modular tensor categories
5. **Prof. Suchitra Sebastian** (Cambridge) — Fractal materials, spectral dimension

We request that [conflicting reviewers] not be selected due to [reason].

Declarations

- This manuscript has not been published elsewhere and is not under consideration at another journal.
- All authors have approved the manuscript and agree with its submission.
- There are no conflicts of interest to declare.
- Supplementary materials include: Python simulation code (available at [repository]), additional figures, and detailed derivations.

Manuscript Details

- Word count: ~8,000 (main text) + ~3,000 (supplementary)

- Figures: 10 (all original, publication-ready)
- References: 87
- Suggested length: Full Article (12 pages + supplementary)

I believe this work will be of significant interest to PRX readers across quantum computing, condensed matter physics, and photonics communities. The combination of theoretical rigor, simulation validation, and concrete experimental roadmap provides a complete pathway from concept to implementation.

Thank you for your consideration. I look forward to your response.

Sincerely,

Ross A. Edwards Independent Researcher Email: ross.edwards@aurphyx.org

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Enclosures:

1. Manuscript (PDF)
2. LaTeX source files
3. Figure files (PDF/PNG)
4. Bibliography (BibTeX)
5. Supplementary Information